Lecture 1 (*February 17*) *Reading: Sections 1.1–2.1*

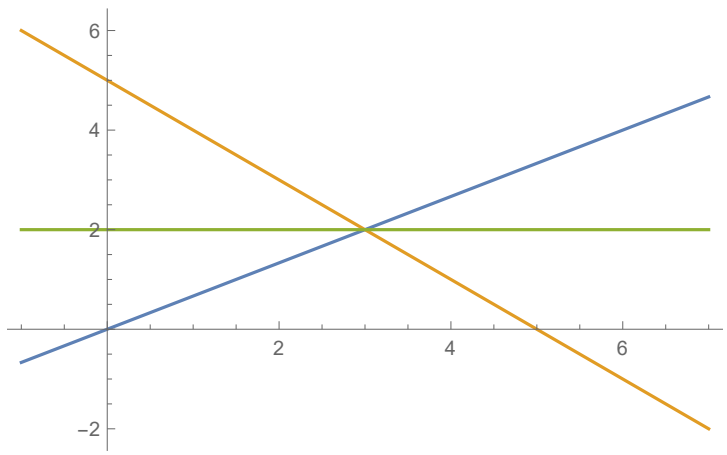
Viewing: Lecture 1

In 18.02, you learned how to solve systems of linear equations:

$$\begin{cases} 2x - 3y = 0 \\ x + y = 5 \\ 2y = 4 \end{cases} \tag{1}$$

The high-school way to solve this system is to take y out from the last equation (since it is the more manageable of the three) and get $y = 2$. Then we plug this answer into either the first or the second equation, and we get $x = 3$. So the solution is $(x, y) = (3, 2)$.

There's also a geometric way to think about the system of equations (1). Linear equations, such as $2x - 3y = 0$, $x + y = 5$ or $2y = 4$, are equations of lines in the xy–plane. Specifically, they correspond to the blue, orange and green lines in the picture below:



Asking for a solution to the system of equations (1) is the same thing as asking for a point (x, y) which lies on all three of these lines, or in other words, the intersection of these lines. It's easy to see from the picture above that this point is $(x, y) = (3, 2)$.

Another point of view on the system of equations (1) is to rewrite it in matrix form:

$$\begin{bmatrix} 2 & -3 \\ 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 4 \end{bmatrix} \tag{2}$$

In this way, you solve for the quantity $\begin{bmatrix} x \\ y \end{bmatrix}$, which is thought of as the point (x, y) in the plane. There are a number of ways to solve matrix equations like (2), the most general form of which is:

$$A\boldsymbol{v} = \boldsymbol{b} \tag{3}$$

where A is an $m \times n$ matrix, $\boldsymbol{v}$ is an $n \times 1$ vector, and $\boldsymbol{b}$ is a $m \times 1$ vector. In situations such as (3), usually A and $\boldsymbol{b}$ are given as part of the problem, while $\boldsymbol{v}$ is the unknown you have to solve for.

Before we delve into ways to solve (3), let us say a thing or two about vectors. We think of:

$$\boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \tag{4}$$

as a point in n–dimensional space. For example, 2–dimensional space is a plane (such as the surface of the paper, or a blackboard), while 3–dimensional space, is... well, ordinary space. Consider the following operations:

- given two vectors $\boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$, you may add them componentwise:

$$\boldsymbol{v} + \boldsymbol{w} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \\ \vdots \\ v_n + w_n \end{bmatrix}$$

- given a vector $\boldsymbol{v}$ and a number α (called scalar), we can multiply them together:

$$\alpha\boldsymbol{v} = \begin{bmatrix} \alpha v_1 \\ \alpha v_2 \\ \vdots \\ \alpha v_n \end{bmatrix}$$

Combining the two operations above, we will say a **linear combination** of two vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ is any expression of the form:

$$\alpha\boldsymbol{v} + \beta\boldsymbol{w} = \begin{bmatrix} \alpha v_1 + \beta w_1 \\ \alpha v_2 + \beta w_2 \\ \vdots \\ \alpha v_n + \beta w_n \end{bmatrix}$$

The numbers α and β are called the **coefficients** of the linear combination. So we've seen the definition, but what does the concept of "linear combination" actually mean? Let's take an easy example:

$$\boldsymbol{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad \boldsymbol{w} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \qquad \Rightarrow \qquad \text{linear combinations are } \alpha\boldsymbol{v} + \beta\boldsymbol{w} = \begin{bmatrix} \alpha \\ 0 \\ \beta \end{bmatrix}$$

As α and β run over all possible numbers, the right-hand side runs over all vectors in the xz–plane (which in turn is a subset of xyz–space), since the y–component of such vectors is 0. Given that $\boldsymbol{v}$ is the unit vector on the x–axis and $\boldsymbol{w}$ is the unit vector on the z–axis, the principle here is the following.

Fact 1. *Almost* [1] *any two* $n \times 1$ *vectors* $\boldsymbol{v}$ *and* $\boldsymbol{w}$ *trace out a 2–dimensional plane in* n*–dimensional space. Any vector in the aforementioned plane is a linear combination of* $\boldsymbol{v}$ *and* $\boldsymbol{w}$*, and vice-versa.*

Example 1. *Let's draw the plane traced out by the vectors* $\boldsymbol{v} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$ *and* $\boldsymbol{w} = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$.

To do this, represent $\boldsymbol{v}$ *and* $\boldsymbol{w}$ *as the points* $(2, 0, -1)$ *and* $(0, 3, 1)$*, respectively, in 3–dimensional space. Then trace out the lines passing through the origin and either the point* $\boldsymbol{v}$ *or* $\boldsymbol{w}$*. Finally, trace out the 2–plane spanned by these two lines.*

We can use the language of linear combinations to make sense of systems of equations such as (3). Let's look at the specific example (2). Because:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ y \end{bmatrix} = x \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

then formula (2) reads:

$$\begin{bmatrix} 2 & -3 \\ 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \cdot \begin{bmatrix} 2 & -3 \\ 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \cdot \begin{bmatrix} 2 & -3 \\ 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = x \cdot \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + y \cdot \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 4 \end{bmatrix}$$

Therefore, solving for (x, y) is the same thing as finding a certain linear combination of the vectors $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}$ (which are nothing but the columns of the 3×2 matrix in the left-hand side of (2)) that produces the vector $\begin{bmatrix} 0 \\ 5 \\ 4 \end{bmatrix}$. Such a linear combination exists if and only if the latter vector lies in the plane spanned out by the former vectors, which is indeed the case in our example. In general:

Fact 2. *The equation* $A\boldsymbol{v} := \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} =: \boldsymbol{b}$ *is equivalent to:*

$$v_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + \dots + v_n \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \tag{5}$$

In other words, solving the system of equations for $\boldsymbol{v}$ *precisely entails finding out which linear combinations of the columns of the matrix* A *are equal to the vector* $\boldsymbol{b}$*.*

[1] The word "almost" requires us to exclude the case when $\boldsymbol{v}$ and $\boldsymbol{w}$ are collinear, i.e. on the same line passing through the origin; we will discuss this issue further when introducing the notion of linear independence

There is another operation that one can perform on two vectors, the **dot product**:

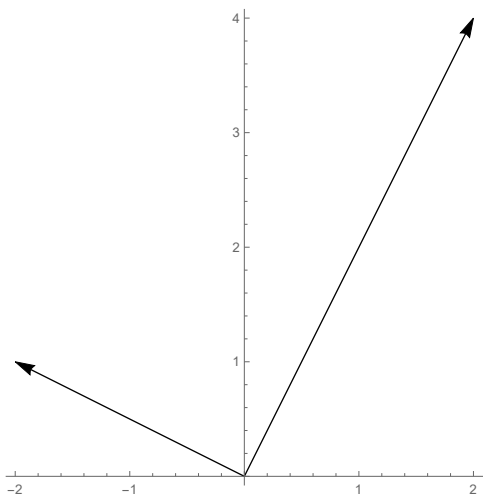
$$\text{if } \boldsymbol{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \quad \text{and} \quad \boldsymbol{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} \quad \text{then} \quad \boxed{\boldsymbol{v} \cdot \boldsymbol{w} = v_1 w_1 + \cdots + v_n w_n}$$

VERY IMPORTANT: the dot product of two vectors is a number (a.k.a. scalar). There is no "reasonable" way to multiply two vectors in such a way as to obtain a vector (with the notable exception of the cross-product for vectors with $n = 3$ components, but that's another story).

The dot product allows us to describe the angle θ between two vectors $\boldsymbol{v}$ and $\boldsymbol{w}$. Recall that:

$$\boldsymbol{v} \cdot \boldsymbol{w} = 0 \qquad \text{means that} \qquad \theta = 90^\circ$$

For example, take $\boldsymbol{v} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$:



The angle between these two vectors is 90°, and indeed $\boldsymbol{v} \cdot \boldsymbol{w} = (-2) \cdot 2 + 1 \cdot 4 = 0$. But the dot product describes arbitrary angles, not just right angles. The general formula is:

$$\boxed{\cos\theta = \frac{\boldsymbol{v} \cdot \boldsymbol{w}}{||\boldsymbol{v}||\ ||\boldsymbol{w}||}} \tag{6}$$

where the **length** of a vector is given by:

$$\boxed{||\boldsymbol{v}|| = \sqrt{\boldsymbol{v} \cdot \boldsymbol{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}} \tag{7}$$

So if you want to compute the angle between two lines, apply formula (6) with $\boldsymbol{v}$ and $\boldsymbol{w}$ being the unit (i.e. length 1) vectors corresponding to those lines. Then the formula in question precisely says that the cosine of the angle is equal to the dot product of unit vectors.